A BiLSTM with Attention Approach Using 2D Skeletons for Autism Activity Recognition

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***Abstract*—Autism is a complex neurodevelopmental disorder that significantly limits the social and cognitive skills of the child, and its symptoms are diverse and difficult to predict. They differ significantly across age ranges and also between males and females, with females tending to hide the autism symptoms, making it more difficult to detect. For this reason, early detection of the autism disorder is challenging through solely therapist or parent intervention, which suggests using automatic detection of the disorder. In this work, we propose a lightweight approach for activity recognition for autism using normalized 2D skeleton data. In contrast to existing methods, our approach reduces model complexity by using only one type of data while preserving high accuracy. It is also aware of data leakage issues and evaluates the model on the MMASD dataset, achieving an accuracy of 94.84%. This performance approaches that of multimodal models, but with fewer computational resources, making it more applicable to real-world scenarios.**

***Index Terms*—Autism, Activity Recognition, 2D Skeleton, Pose Estimation, Lightweight Deep Learning**

I. Introduction

Autism Spectrum Disorder (ASD) is a neurodevelopmental disorder characterized by challenges in social interaction, communication, and cognitive tasks. These challenges often appear through repetitive behaviors or weak engagement and attention.

For this reason, early detection of autism is critical for multiple reasons, as outlined in [1]. These include more effective intervention; helping the child's family to understand and respond to their needs sooner; improvement of therapy outcomes; reduction of costs resulting from delayed detection; and personalization of educational approaches at an earlier stage.

Autism is a very challenging disorder as its symptoms vary widely across different age ranges and between genders. It is more difficult to detect autism in females than in males, as females tend to hide their emotions, as mentioned in [2].

Therefore, depending solely on therapists or parents for early diagnosis may not be reliable, as they may not recognize or accurately estimate the child's autistic behaviors, especially those that are not obvious. This highlights the need for more intelligent diagnosis approaches utilizing artificial intelligence.

Several studies have applied AI for activity recognition in autistic children using datasets such as SSBD, DREAM, and MMASD. Lokegaonkar et al. [3] extended SSBD with more videos and used a ResNet-18 and Bi-LSTM, achieving 81% accuracy. Saha et al. [4] classified tasks in DREAM with 3D skeleton data, reaching 82.01% accuracy. Washington et al. [5] used OpenPose keypoints and a CNN-LSTM, achieving an F1-score of 90.77% on SSBD. Shin et al. [6] combined a ConvNeXt-Base CNN and MSG3D on DREAM, obtaining 70.19% accuracy for ASD severity. Singh et al. [7] expanded SSBD to ESSBD with new behaviors, achieving 92.6% accuracy using an LRCN. Jeyarani and Senthilkumar [8] proposed DSVTN-ASD, a self-supervised transformer on ESSBD with RTMPose keypoints, achieving a macro AUROC of 93.13% but with high computational cost. Several of the previously discussed works advanced autism activity recognition, but few considered model complexity—crucial for real-world deployment.

In this paper, we tackle the problem of autism activity recognition by using a lightweight deep learning approach based on 2D skeleton data and a BiLSTM with multi-head attention, achieving an accuracy of 94.84%. To this end, we worked on the MMASD dataset [9], which is the most recent and comprehensive dataset for autism activity recognition. MMASD consists of recorded therapy sessions and includes 11 activities across three categories: robotic, music, and yoga. It provides four modalities: 2D skeleton, 3D skeleton, optical flow, and clinical evaluation scores. We applied a BiLSTM with multi-head attention to the 2D skeleton data, after performing custom normalization using relative joint positions with respect to the mid-hip joint, and used session-based splitting to prevent data leakage. Our model demonstrates that efficient processing of a single modality from a multimodal dataset can still yield high

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accuracy while significantly reducing computational cost.

The rest of the paper is organized as follows: Section II covers related work; Section III details the methodology; Section IV presents the results; and Section V concludes the paper.

II. Related Work

Three key public datasets support autism activity recognition: SSBD, DREAM, and MMASD. SSBD [10] includes 75 videos of stimming behaviors, such as head banging and arm flapping. DREAM [11] contains imitation and joint attention tasks recorded in robot- and human-led sessions. **MMASD** [9] provides 11 therapy-based activities, including Body Swing and Sing and Clap. These datasets have supported several studies on autism activity recognition using diverse models.

A. Vision-Based Approaches for Activity Recognition

Wei et al. [12] presented a vision-based approach for autism activity recognition by extending the SSBD dataset. They added new videos collected from platforms such as YouTube, Vimeo, and Dailymotion, while removing noisy samples to improve data quality. Their method uses an Inflated 3D ConvNet (I3D) to extract spatio-temporal features from short RGB video clips and employs a Multi-Stage Temporal Convolutional Network (MS-TCN) for temporal modeling, achieving a weighted F1-score of 0.83. Although this approach achieves high accuracy and enriches the existing dataset, it relies on processing raw video frames, which can be computationally intensive for real-time or resource-constrained deployment.

B. Vision-Language Approaches for Autism Behavior Recognition

Recent work has explored vision–language models for autism behavior recognition. Deng et al. [13] proposed using textual descriptions of behavior classes (e.g., *head banging*, *arm flapping*) to guide training. They used the SSBD dataset and its extension, ESBD, which adds a fourth class, *hand action*. Their model combines a Video Swin Transformer for visual feature extraction and a text encoder for language feature embedding. This approach achieved 85.01% accuracy on SSBD and 88.17% on ESBD, outperforming vision-only baselines. However, its high computational cost and dependence on high-resolution video limit real-time deployment.

C. Activity Recognition Using 2D Skeletons

Kanwal et al. presented "ARAIS—Activity Recognition of Autism Children During Intervention Sessions" [14], using the MMASD dataset. They utilized only 2D skeleton data, normalized by dividing x and y coordinates by 640 and 480, respectively. Their model, comprising two LSTM layers (128 units each), a fully connected layer, and a SoftMax classifier, achieved 95.72% accuracy using a random data split without session split awareness.

D. Activity Recognition Using Multimodal Data

Another paper that utilized the MMASD dataset is by Ravva *et al.*, titled "MMASD+: A Novel Dataset for Privacy-Preserving Behavior Analysis of Children with Autism Spectrum Disorder" [15]. This work extends the original MMASD dataset by extracting a new modality: 3D body mesh using the ROMP model. The authors employ YOLOv8 and Deep SORT to distinguish between the child and therapist in the video frames. Then, they apply a Video Vision Transformer (ViViT) for the 3D mesh data, a 3D CNN for optical flow, and an LSTM for 3D skeleton sequences. Multi-head attention and fusion layers are used to integrate features across the three modalities. The model achieves 96.8% accuracy for activity classification and 95.3% accuracy for ASD classification.

E. Modality Complexity Comparison

While multimodal approaches have achieved the highest accuracy, they come with higher cost, which can be impractical for real implementation. For this, we propose to process one type of modality, which is 2D skeleton, as it is the least resource-intensive among the four modalities, but with session split awareness for preventing data leakage. The modality complexities are summarized in TABLE I, with 2D skeletons being the simplest, using only x and y coordinates for 25 joints.

TABLE I
Comparison of Modality Complexity

Modality	**Complexity Level**	**Complexity Description**
2D Skeleton	Low	25 joints; includes only x and y coordinates (no depth)
3D Skeleton	Moderate	71 joints; includes x, y, and z coordinates
Optical Flow	High	Pixel-level motion estimation; requires dense computation
3D Mesh	Very High	Full-body surface model with 6,890 vertices

III. Methodology

In this section, we discuss the detailed method used for activity recognition of autistic children using 2D skeletons.

A. MMASD Dataset

MMASD [9] is the most recent, comprehensive, and diverse multimodal public dataset. The data was recorded during therapist-led sessions involving 32 children with autism, aged between 5 and 12 years. The children participated in over 100 hours of play sessions, which were segmented into 1,315 video clips covering 11 activities across three categories: robotic (Arm Swing, Body Swing, Chest Expansion, Squat), music (Drumming, Maracas Forward Shaking, Maracas Shaking, Sing and Clap), and yoga (Frog Pose, Tree Pose, Twist Pose). The dataset provides multiple modalities, including 2D skeletons, 3D skeletons, optical flow, and clinical evaluation scores.

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B. Relative Normalization of Skeleton

Variations in body size, camera position, and session conditions can significantly affect the learning process. For example, if one child appears on the right side of the frame and another on the left, even if they perform the same activity, the model may misclassify them due to differences in spatial position. This highlights the importance of proper normalization when preparing data for activity recognition.

To address this, we apply relative normalization. First, we center each skeleton by subtracting the coordinates of the mid-hip joint (considered the center of the body) from all other joints. This makes all skeletons aligned to a common reference point — the mid-hip joint.

Second, we scale each skeleton by dividing all joint positions by the Euclidean distance between the left and right shoulder joints. This represents the width of the individual and helps normalize differences in scale due to body size or distance from the camera. As a result, we obtain standardized skeleton sizes relative to each individual.

As illustrated in Figure 1, these two steps enable the model to use relative rather than absolute coordinates, enhancing robustness across subjects and sessions.

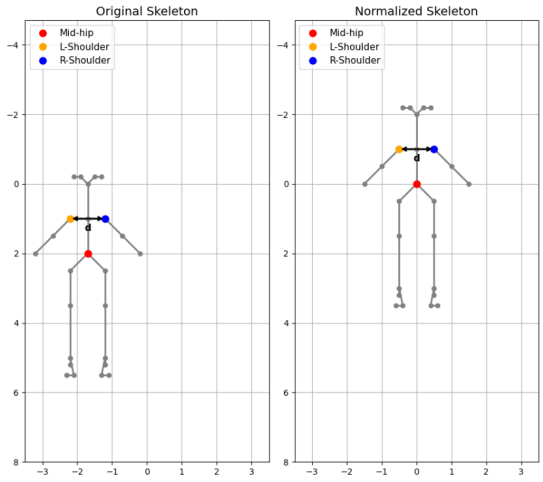


Fig. 1. The original skeleton (left) is centered on the mid-hip joint and scaled using the distance between the shoulders to produce a standardized pose (right).

C. Model Pipeline

The model pipeline, summarized in Table II, is as follows:

- We forward the normalized skeleton data into a BiLSTM layer. Each input is a sequence of 60 frames, and each frame contains 25 keypoints with x and y coordinates, resulting in 50 features per frame.
- The BiLSTM layer, which captures temporal dependencies in both forward and backward directions, consists of 2 layers with 128 hidden units each.
- After that, we apply a multi-head attention mechanism with 4 heads to focus on the most informative parts of the sequence.
- A residual connection is then added to the attention output and followed by a Layer Normalization step to stabilize training and improve gradient flow.
- We apply temporal mean pooling by averaging across the time dimension to reduce the sequence into a fixed-size representation.
- Then, we use a projection layer to reduce the dimensionality of the output from 256 to 64, followed by a ReLU activation.
- A dropout layer with a rate of 0.3 is applied to prevent overfitting.
- Finally, a fully connected layer maps the 64-dimensional vector to 10 output classes, each representing one activity.

TABLE II
MODEL SUMMARY

Layer	Input Shape	Output Shape
Input (Skeleton Sequence)	(60, 50)	(60, 50)
BiLSTM (2 layers, 128 units)	(60, 50)	(60, 256)
Multi-Head Attention (4 heads)	(60, 256)	(60, 256)
Residual + LayerNorm	(60, 256)	(60, 256)
Temporal Mean Pooling	(60, 256)	(256)
Projection (Linear + ReLU)	(256)	(64)
Dropout (p=0.3)	(64)	(64)
Fully Connected (Softmax)	(64)	(10)

D. Training

We grouped frames by session ID to prevent data leakage, then split the dataset into 80% training and 20% validation. The model, shown in Fig. 2, was trained on a GPU for 100 epochs with early stopping (patience of 10), using a batch size of 32, learning rate of 0.001, cross-entropy loss, and the Adam optimizer.

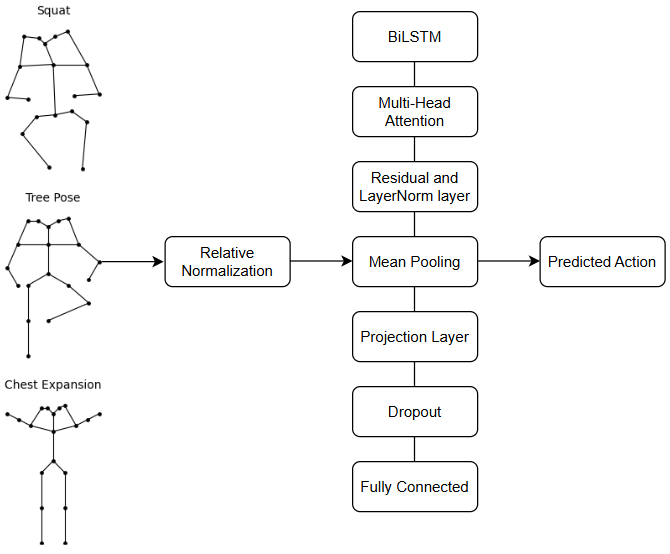


Fig. 2. Model Architecture

IV. RESULTS

In this section, we discuss the results obtained by our proposed model.

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A. Overall Performance

Our BiLSTM with attention model achieved a validation accuracy of **94.84%** using only 2D skeleton data from the MMASD dataset [9]. The training was based on a session-based split (80% training, 20% validation) to ensure no overlap between sessions, preventing data leakage and providing realistic results, as summarized in TABLE III.

The model converged quickly, with the training loss dropping significantly within the first 20 epochs, as shown in Fig. 3. Validation accuracy peaked at epoch 11 and remained stable through early stopping at epoch 21, as illustrated in Fig. 4.

The model approaches the performance of multimodal models while using only a single modality (2D skeleton), thereby reducing computation while maintaining high accuracy.

TABLE III
PERFORMANCE METRICS

Metric	Score
Validation Accuracy	94.84%
Macro Precision	94%
Macro Recall	94%
Macro F1-score	94%

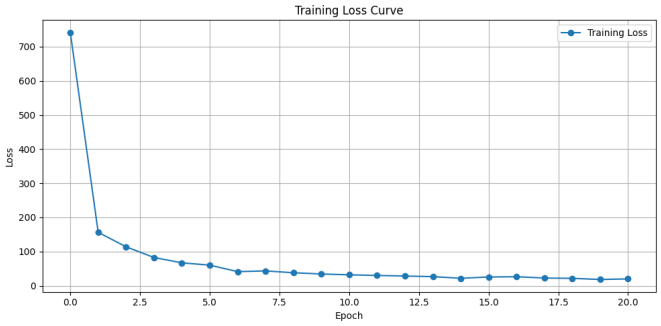


Fig. 3. Training loss over epochs. The model converges quickly, with the loss decreasing significantly before early stopping at epoch 21.

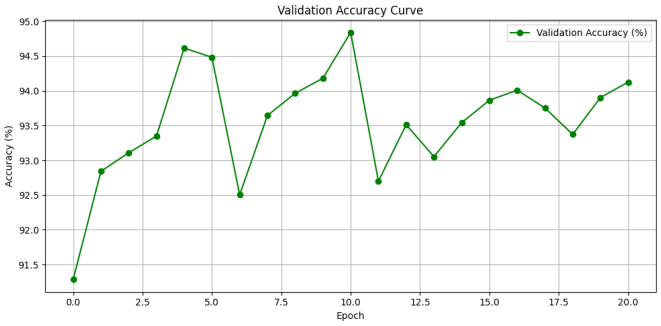


Fig. 4. Validation accuracy over epochs. Accuracy stabilizes around epoch 11 and remains consistent until early stopping.

B. Confusion Matrix Analysis

To better understand the model performance, we analyzed the confusion matrix, as shown in Fig. 5. The model achieved strong performance across most classes, with a high number of correct predictions along the diagonal. However, a few activity pairs showed overlapping due to visual or motion similarity.

- **Body Swing (bs)** was misclassified as **Chest Expansion (ce)** in 5.36% of cases. Both activities involve repetitive upper-body motions, making their temporal patterns hard to distinguish.
- **Maracas Shaking (ms)** was misclassified as **Maracas Forward Shaking (mfs)** in 9.42% of cases. The actions are nearly identical in arm movement, with only a subtle difference in direction, which is difficult to capture in 2D skeleton data.
- **Sing and Clap (sac)** was misclassified as **Drumming (dr)** in 20.39% of cases. The static poses of both activities often appear similar, with arms raised near the chest, leading to visual confusion in skeleton representation.

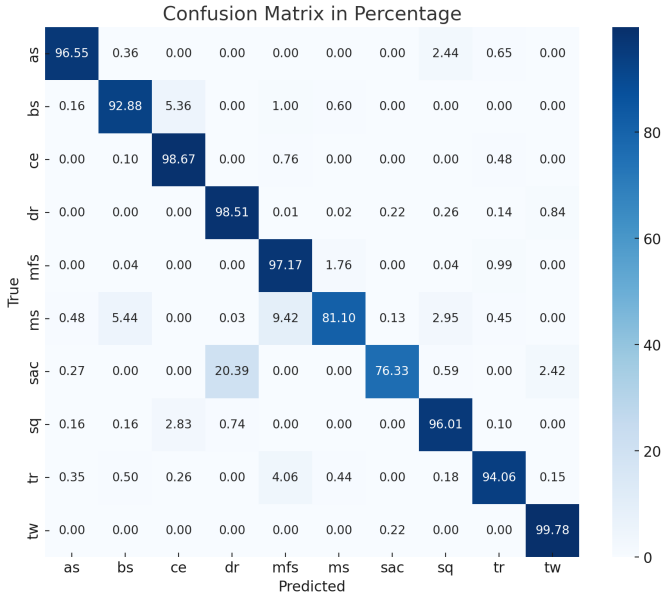


Fig. 5. Confusion matrix showing model predictions across the 10 activity classes. Most activities are correctly predicted, with some overlap between visually or motionally similar classes.

C. Discussion

Comparing our model with previous models that utilized the MMASD dataset, such as ARAIS [14] and MMASD+ [15], we can observe that our model achieved an accuracy close to those reported in the literature. A summary of these comparisons is provided in TABLE IV. The ARAIS model achieved a slightly higher accuracy of 95.72%, but it used a random data split without session-awareness, which may introduce data leakage and lead to overestimated performance.

On the other hand, MMASD+ achieved a stronger accuracy of 96.8% by fusing three complex modalities: 3D skeleton, optical flow, and 3D body mesh. However, this comes with significantly higher computational requirements.

Although our model achieved a slightly lower accuracy, it uses only a single, lightweight modality — the 2D skeleton

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— making it more feasible for real-world applications where computational resources may be limited. It represents a good trade-off between performance and complexity.

While our proposed model shows strong overall performance, it also highlights areas where the model can still improve, particularly in distinguishing between similar activities.

TABLE IV
COMPARISON OF ACTIVITY RECOGNITION MODELS ON MMASD DATASET

Paper	Modality Used	Model	Accuracy	Data Split
ARAIS	2D Skeleton	LSTM	95.72%	Random Split
ARAIS (re-implemented)	2D Skeleton	LSTM	83.13%	Session Split
MMASD+	3D Mesh, Optical Flow, 3D Skeleton	LSTM + 3D CNN + ViViT + Multi-head Attention	96.8%	Session Split
Ours	2D Skeleton	Relative Normalization + BiLSTM + Multi-head Attention	94.84%	Session Split

V. CONCLUSION

In this work, we proposed an efficient activity recognition model that uses only 2D skeleton data, which we normalized by centering each skeleton on the mid-hip joint and scaling it based on shoulder distance. The normalized sequences were then processed by a BiLSTM network with multi-head attention to capture important temporal dependencies. Using this lightweight approach, our model achieved an accuracy of 94.84%, comparable to more complex multimodal methods but with significantly lower computational cost, demonstrating that optimizing a single modality can provide both high performance and practical efficiency.

However, at the same time, further improvement is still required to optimize the classification of similar activities, as the model still faces challenges distinguishing between activities with highly similar motion patterns, such as chest expansion and clapping actions. Addressing such challenges may require incorporating additional temporal features, such as joint velocities, or employing more advanced pose estimation techniques.

In future work, we plan to test our model with autistic children in real-world scenarios to evaluate its practical performance and usability in therapy sessions.

Overall, this work demonstrates that careful processing of 2D skeleton data combined with an efficient temporal model can help balance the trade-off between accuracy and computational complexity for autism activity recognition.

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